

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 236 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	D And F Block Elements - Core Application	236 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 D And F Block Elements - Core Application (Very High Yield — 236 PYQs)

Core Principle: Key problem pattern in D And F Block Elements.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : The number of pairs among $[Ti^{4+}, V^{2+}], [V^{2+}, Mn^{2+}], [Mn^{2+}, Fe^{3+}]$ and $[V^{2+}], \dots$ " [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "A paper dipped in a dil. H_2SO_4 solution of 'X' upon treatment with SO_2 gas turns into green. The compound 'X' is :..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "The correct statements among the following are, A. $Mo(VI)$ and $W(VI)$ are less stable than $Cr(VI)$. B. Ce^{4+} and Tb^{4+} are oxida..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Given below are two statements : Statement I : The number of pairs among $[\text{Ti}^{4+}, \text{V}^{2+}]$, $[\text{V}^{2+}, \text{Mn}^{2+}]$, $[\text{Mn}^{2+}, \text{Fe}^{3+}]$ and $[\text{V}^{2+}, \text{Cr}^{2+}]$ in which both ions are coloured is 3 . Statement II : The number of pairs among $[\text{La}^{3+}, \text{Yb}^{2+}]$, $[\text{Lu}^{3+}, \text{Ce}^{4+}]$ and $[\text{Ac}^{3+}, \text{Lr}^{3+}]$ ions in which both are diamagnetic is 3 . In the light of the above statements, choose the correct from the options given below :

- (A) Both Statement I and Statement II are correct **[Correct]**
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Correct Answer: A

Solution Analysis: Coloured ions have unpaired electron. (i) $\text{Ti}^{4+} = 4s^0 3d^0$ $n=0$ (ii) $\text{V}^{2+} = 4s^0 3d^3$ $n=3$ (iii) $\text{Mn}^{2+} = 4s^0 3d^5$ $n=5$ (iv) $\text{Fe}^{3+} = 4s^0 3d^5$ $n=5$ (v) $\text{Cr}^{2+} = 4s^0 3d^4$ $n=4$ The absence of unpaired electron makes the element diamagnetic.
$$\begin{aligned} & \text{La}^{3+} = [\text{Xe}] 5d^0 6s^0 \quad \& \quad 4b^{2+} = [\text{Xe}] 4f^{14} 6s^0 \quad \& \\ & \text{Lu}^{3+} = [\text{Xe}] 4f^{14} 5d^0 6s^0 \quad \& \quad \text{Ce}^{4+} = [\text{Xe}] 4f^0 5d^0 6s^0 \quad \& \quad \text{Ac}^{3+} = [\text{Rn}] 6d^0 7s^0 \end{aligned}$$